CURRICULUM VITAE

Wookyoung Kim

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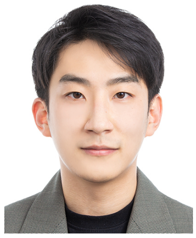

EDUCATION

Ph.D. KAIST, Department of Mechanical Engineering (2021)
Advisor: Sung Jin Kim
M.S. KAIST, Department of Mechanical Engineering (2017)
B.S. KAIST, Department of Mechanical Engineering (2015)

RESEARCH INTERESTS

- **Data Center Cooling**: Immersion cooling / Direct liquid cooling / Jet impingement / PUE optimization
- **High-Heat-Flux Electronics Cooling**: CPU/GPU cold plate / Focused cooling / Next-gen chip thermal management
- **Heat Exchangers**: PCHE / Shell & Tube / Plate HX / Cryogenic applications
- **Heat Pump Systems**: Chemisorption / Adsorption / High-temperature (300°C)
- **Low GWP Refrigerant Systems**: Eco-friendly refrigerants / VLE measurement / Equation of state development
- **Two-Phase Heat Transfer**: Boiling / Evaporation / Pulsating heat pipes / Vapor chambers
- **Cryogenic Heat Transfer**: Liquid hydrogen vaporizer / Supercritical fluid / Sub-200°C systems

RESEARCH OUTPUTS

- **SCI/SCIE Journal Papers:** 11
- **KCI Journal Papers:** 8
- **Conference Presentations:** 11
- **Domestic Patents (registered):** 18
- **U.S. Patents (registered):** 1
- **Registered Software Programs:** 8
- **Technology Transfers to Industry:** 5
- **Research Projects (PI/Lead / Total):** 8 / 20

SELECTED PUBLICATIONS

(1) **W. Kim** et al., "Freezing phenomenon in PCHE for cryogenic LH2 vaporizer," *Appl. Therm. Eng.* 273 (2025).
(2) **W. Kim** and S.J. Kim, "Fundamental issues about pulsating heat pipes," *J. Heat Transfer - ASME* 143 (2021).
(3) **W. Kim** and S.J. Kim, "Flow behavior effect on pulsating heat pipes," *Int. J. Heat Mass Transfer* 149 (2020).
(4) **W. Kim** and S.J. Kim, "Reentrant cavities on pulsating heat pipe," *Appl. Therm. Eng.* 133 (2018).
(5) **J.S. Kim, W. Kim** et al., "Pool boiling of ammonia outside enhanced tubes," *Appl. Therm. Eng.* 247 (2024).

RESEARCH PROJECTS (PI / Lead)

Showing 8 of 20 total projects (PI/Lead role).
(1) **Development of an AI-based Automated Design Tool for Data Center Direct Liquid Cooling (DLC) Systems** (PI) [2026]
(2) **Development of a Direct Liquid Cooling (DLC) System for Reducing Power Consumption in Data Centers** (Lead) [2026-2029]
(3) **Jet-Enhanced Immersion Cooling for Data Centers** (PI) [2025]
(4) **KIMMPROP - Thermophysical Property iOS/Android App** (PI) [2025]
(5) **PCHE for Liquid Hydrogen Supply System** (Lead) [2021-2026]
(6) **Compact PCHE Design for Liquid Hydrogen** (Lead) [2022-2026]
(7) **Immersion Cooling Waste Heat Utilization** (Lead) [2024-2028]
(8) **PCHE Freezing Test for Vehicle Heat Exchangers** (PI) [2025]

PROFESSIONAL ACTIVITIES

(1) **Council Member**, Korean Society of Mechanical Engineers (KSME) [2026.04-Present]
(2) **Committee Member - Thermal Management / Cooling**, SAREK Data Center Technology Committee [2026.03-Present]

(3) **Regular Member**, Society of Air-Conditioning and Refrigerating Engineers of Korea (SAREK) [2022.04-Present]

(4) **Regular Member**, Korean Society of Mechanical Engineers (KSME) [2021.05-Present]